Babisan Pirapakaran

647-739-4626 | babisanp@gmail.com | linkedin.com/in/babisan-p | github.com/NotTolerable

Education

University of Waterloo Waterloo, ON

BMath, Computational Mathematics - Computer Science Minor *Expected 2029*

- Coursework: Data Structures & Algorithms, Linux, Shell Scripting, Pre-processing, Computer Architecture, Low-Level Systems, Assembly Language Programming, Memory Hierarchy & Caching

Experience

Software Engineer Intern May 2026 – Present

407 ETR — Platforms and Infrastructure Team *Vaughan, ON*

- Owned AWS S3 configuration for the migration of **200M+** historical customer invoices, building a Python tool that generated **160** lifecycle policies to automate archival and retention-based expiration.
- Automated Helm chart generation, reducing Kubernetes deployment setup time from **4+ hours** to **15 minutes.**
- Resolved **2** critical security vulnerabilities blocking deployment across Jenkins CI/CD pipelines and OpenShift builds for a developer platform used by **50+ developers**, restoring blocked deployments.
- Provisioned **40+** AWS and OpenShift resources for a Temporal POC using Terraform and Terragrunt.
- Enhanced an internal Java Spring Boot and Solace queue-visualization platform used by **100+ engineers**, adding queue-message retrieval and download functionality through REST APIs and testing integrations with Postman.
- Investigated and traced delivery paths for **10+** customer email-delivery incidents using AWS SES and CloudTrail.

Software Engineer Sept 2024 – Aug 2025

DevRoom

- Operated in Linux Ubuntu production environments across **20+** client deployments, improving release reliability.
- Built backend infrastructure for a network spanning across proxies, hub services, and 8 service stacks for **300** users.
- Resolved an authentication and data-integrity incident affecting **550** users and **1000+** records.

Software Developer May 2023 – Apr 2024

Orbital Studios

- Automated CI/CD workflows with GitHub Actions and Maven across nine repositories.
- Built multi-threaded backend services supporting **200+** concurrent users across 5 service stacks.
- Tuned JVM heap/GC settings, established baseline performance checks, reducing crashes and instability by **80%**.

Technical Skills

Languages: C++, Python, Java, SQL, RISC-V Assembly, JavaScript, TypeScript, C, HTML/CSS
Frameworks: Spring Boot, FastAPI, Flask, React, Next.js, Vite
Cloud/DevOps: AWS (S3, SES, CloudTrail), GCP, Docker, Kubernetes, Red Hat OpenShift, Terraform, Terragrunt, Helm, Jenkins, Postman, Github Actions, Solace, Podman, Supabase, Linux (Ubuntu)

Projects

Axiom – Low-Latency Multi-Venue Trading System | *C++20, Linux, DPDK, FIX/SBE, CMake, Prometheus*

- Built a C++20 execution system to normalize market data and order entry across crypto and prediction-market venues, maintaining in-memory order books and routing orders by price, liquidity, fees, and venue health.
- Optimized the event-to-order hot path using DPDK kernel bypass, CPU pinning, huge pages, preallocated memory pools, and lock-free ring buffers; added pre-trade risk checks, deterministic market replay, and p50/p99 latency benchmarks.

Verilly – AI Compliance Verification Engine | *Python, FastAPI, LangGraph, Pydantic, Next.js*

- Built a LangGraph workflow that extracts facts from technical documents and maps enterprise questionnaires to security and AI governance controls using schema-constrained LLM outputs.
- Created a versioned dependency graph to model relationships between security controls and their prerequisites.
- Developed a deterministic verification engine to block unsupported claims and generate cited compliance reports.

Behaviourly – AI Interview & Behavioral Analysis Platform | *React, FastAPI, PostgreSQL, Supabase, Presage*

- Built a multimodal interview-analysis pipeline using Presage to process video frames for behavioral signals, OpenAI's Whisper for speech transcription, and Gemini API for transcript-based response evaluation.
- Developed FastAPI services to combine visual metrics, transcripts and role-specific interview context into timestamped feedback, weak-moment clips, and performance scores.